

CIRCUMFERENCE

Lesson 3: Circumference Rounded Out

Australian Curriculum: Mathematics - Year 8

ACMMG197: Investigate the relationship between features of circles such as circumference, area, radius and diameter. Use formulas to solve problems involving circumference and area.

Lesson abstract

This lesson provides a set of consolidation, application and extension activities. Activities include estimation and calculation of the sizes of real world circular objects; solving a famous puzzle numerically or algebraically; and using the formula in multi-step calculations, and an open design challenge.

Mathematical purpose (for students)

To use the formula for circumference of a circle in simple and complex situations.

Mathematical purpose (for teachers)

The lesson focus is to build students' skills with computations involving circumference - both to calculate circumference from the diameter or radius and to calculate the diameter and/or radius from a given circumference. Measurements show a range of number types- whole numbers, decimals with differing numbers of decimal places and fractions, both proper and mixed numbers. Students consider how accuracy of answers is limited by the accuracy of given measurements. The lesson provides a task that could be used for assessment.

At the end of these activities, students will be able to:

- Present calculations in a form that can be readily followed by others;
- Check by estimation whether answers are reasonable.

Lesson Length 60-120 minutes approximately

Vocabulary Encountered

- formula
- substitute
- decimal places
- perimeter
- compound shape

Lesson Materials

- scientific calculators
- images for pie and real world estimation/calculation (3a Circumference Task PowerPoint)
- [Student Sheet 1 - Circumference Fluency](#) (1 per student, optional)
- [Teacher Sheet 1 - Circumference Fluency](#)
- [Student Sheet 2 - Jewellery Design](#) (1 per student, optional)
- [Student Sheet 3 - Open Jewellery Design](#) (1 per student, optional)

We value your feedback after this lesson via <http://tiny.cc/lesson-feedback>



Getting Started

The image ‘What is Wrong with this Pi?’ (3a Circumference Task PowerPoint) is a quick fun ‘spot the error’ exercise. Its purpose is to remind students that π is a symbol for the ratio of the circumference of a circle to its diameter and is approximately 3.142 (rounded from 3.14159...).



Initial Task

Using the 3a Circumference Task PowerPoint, and using the formula $C = \pi d$, students decide disputed answers. Move at a speed that allows student to do the calculations, and then check their responses against those on the final slide in each group of four:

- Slide 1 describes a scenario.
- Slide 2 provides limited visual information.
- Slide 3 provides a frame of reference.
- Slide 4 shows the calculated measurement.

Students use their scientific calculators to find the requested values using $C = \pi d$, before final voting on whether the calculated value is less than or greater than the given value. It will be necessary at times to discuss what calculation is appropriate.

Teacher Notes

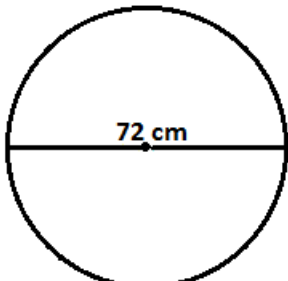
- The first images are those that were used early in Lesson 1, and some extra have been provided so choices can be made.
- Draw the attention of students to an acceptable presentation of the calculations.

Circumference Fluency Task

The optional [Student Sheet 1 - Circumference Fluency](#) provides a variety of scenarios that develop fluency with calculations involving circumference. Students should present their calculations in a form that can be easily followed by another year 8 student.

Solutions should include:

- A diagram showing measurements given.
- The appropriate formula.
- The value to be substituted.
- The result of the calculation.
- The answer to the question with units.



$$C = \pi d$$

When $d = 72$,

$$C = \pi \times 72$$
$$\approx 226$$

The circumference is approximately 226 cm.

Students should be encouraged to check the reasonableness of their answers. For the example above, the circumference should be a bit bigger than 3×70 , say 220 cm.

Students should be encouraged to carefully monitor the correctness of the answers, to build self-assessment skills, and take responsibility for their learning.

Rope around the Earth

Students use their knowledge of circles to find a very surprising result.

Main task

A rope is placed around the **Earth**, at the equator, so it fits snugly like a belt.

- If the rope is then raised 1m off the ground all around the earth, how much extra rope is needed so the rope stays connected all the way around? (ANS: 6.28m)
- If an extra 20 m of rope is added to the original rope, what is the largest object (bacteria, chicken, human, truck, skyscraper) that could now pass under the rope? (ANS: the height of the longer rope is 3.18m above the surface)

What about a rope around a different object:

- A **20 cent piece**?
- Our **sun**?

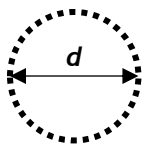
Does the size of the original object matter? Can you justify your answer?

Teacher Notes

First students should guess the answers. Then allow students time to develop their own strategies. They will probably want to work with numbers first, rather than algebra.

- Earth diameter = 12 742 km
- Sun diameter = 1 392 000 km
- 20 cent piece diameter = 28.52 mm

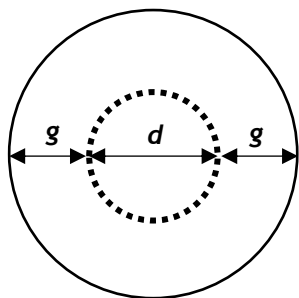
Students should be encouraged to justify algebraically that the height of the longer rope above the surface is independent of the size of the original object.



Algebraic Solution: The size of the original circle does not matter.

Little circle (rope on the ground - dotted)

$$\text{diameter} = d \quad \rightarrow \quad C_{\text{Little}} = \pi d$$



Bigger circle (rope raised g metres off the ground)...

$$\text{diameter} = d + 2g \quad \rightarrow \quad C_{\text{Bigger}} = \pi(d + 2g)$$

$$\text{Also, we know } C_{\text{Little}} + 20 = C_{\text{Bigger}}$$

$$\text{Therefore } \pi d + 20 = \pi(d + 2g)$$

$$\pi d + 20 = \pi d + 2\pi g$$

$$20 = 2\pi g$$

$$g = \frac{20}{2\pi} \approx 3.18 \text{ m}$$

Regardless of the diameter of the object, if the length of the circumference is increased by 20m the radius of the original object is increased by approximately 3.18m. An adult and a medium-sized truck could both pass under it.

Design Applications and Possible Assessment

Jewellery Design

[Student Sheet 2 - Jewellery Design](#) is suitable to use for consolidation or assessment of direct calculations (radius/diameter to circumference). It involves multi-step calculations, and some use of simple geometry. A rubric is provided, which could be given to students if the task is to be used for assessment.

Answers:

Metal cost: \$2073.45; \$2528.98; \$2261.95; \$2073.45.

Bonus question: The bonus question is to explain why Designs 1 and 4 use equal lengths of wire. Some students may provide an algebraic answer or a general argument expressed fully in words. Others may show that the numerical values of the components are the same.

Algebraic solution:

The fourth design is made of one large circle, two semicircles of half its diameter, four semicircles of one quarter its diameter and eight circles of one eighth of its diameter. Let the diameter of the large circle be d . Then the length of metal in the fourth design is $\pi d + 2 * \left(\frac{1}{2}\right) \pi \left(\frac{d}{2}\right) + 4 * \left(\frac{1}{2}\right) \pi \left(\frac{d}{4}\right) + 8 * \pi \left(\frac{d}{8}\right) = 3\pi d$, which is three times the diameter of the large circle.

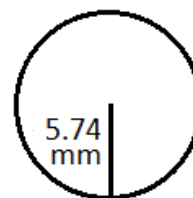
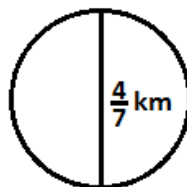
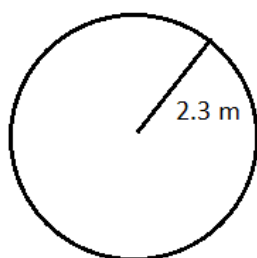
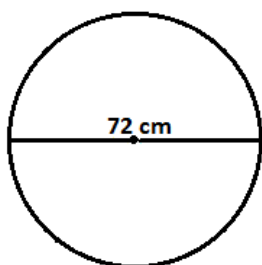
Jewellery design assessment rubric

CRITERION	SCORE
Use of the correct formula, $C = \pi d$ or $C = 2\pi r$	1 2 3 4
The correct calculation for one circle in diagrams 1 and 2	1 2
The correct calculations for all circles in diagrams 1 and 2	1 2
Partial correct reasoning or calculation for diagram 3	1
Complete correct reasoning or calculation for diagram 3	1
Correct calculations for all circles in diagram 4	2
Circumference x 25 for costings for each piece, with allowance for profit	1 2 3 4
Clear and logical presentation of thinking, calculations and price estimate	1 2 3 4
TOTAL 20	
BONUS QUESTION	

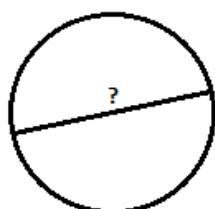
Open Jewellery Design

[Student Sheet 3 - Open Jewellery Design](#) allows students to use their creativity while meeting multiple criteria for the design.

1. Find the **circumference** of each of these circles. *These items vary number type and radius/diameter.*



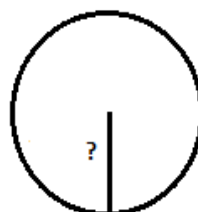
2. Find the **diameter/radius** of each of these circles. *These items vary both number type and radius/diameter.*



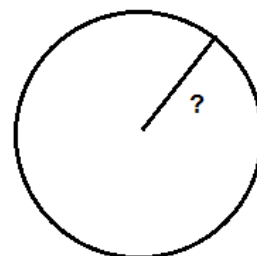
$C = 3.142 \text{ m}$



$C = 4 \text{ m}$

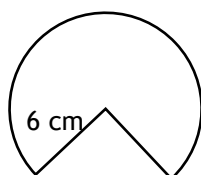
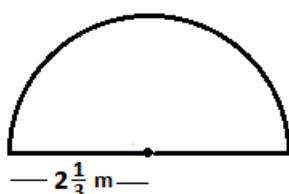


$C = 22.8 \text{ cm}$

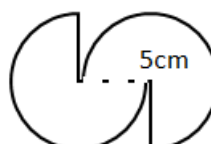


$C = (2x + 4) \text{ m}$

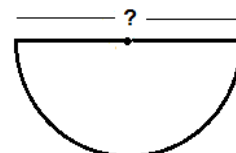
3. Find the **perimeter** of these shapes that are sectors (fractions) of circles. *These are multi-step problems.*



What assumptions are needed? Why?



What assumptions are needed? Why?



Length of curve = 15.23 m

Teacher Sheet 1 - Circumference Fluency

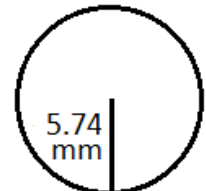
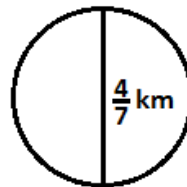
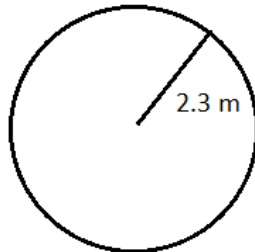
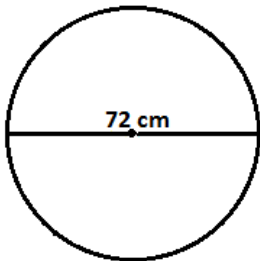
1. Find the **circumference** of each of these circles. *These items vary number type and radius/diameter.*

$$C = 226.19 \text{ cm}$$

$$C = 14.45 \text{ m}$$

$$C = 1.80 \text{ km}$$

$$C = 36.07 \text{ mm}$$



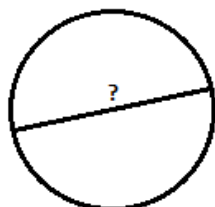
2. Find the **diameter/radius** of each of these circles. *These items vary both number type and radius/diameter.*

$$d = 1.00 \text{ m}$$

$$d = 1.27 \text{ m}$$

$$r = 3.63 \text{ cm}$$

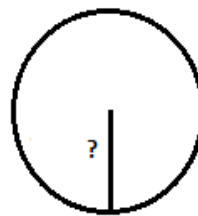
$$r = \frac{(2x+4)}{2\pi} = \frac{(x+2)}{\pi}$$



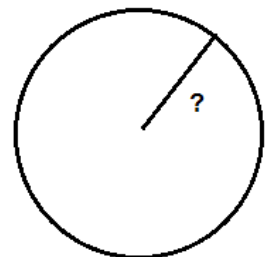
$$C = 3.142 \text{ m}$$



$$C = 4 \text{ m}$$



$$C = 22.8 \text{ cm}$$



$$C = (2x + 4) \text{ m}$$

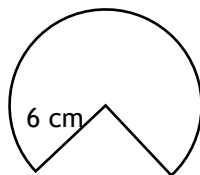
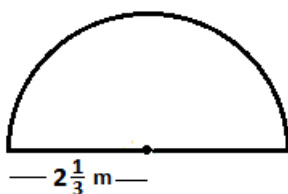
3. Find the **perimeter** of these shapes that are sectors (fractions) of circles. *These are multi-step problems.*

$$P = 12.00 \text{ m}$$

$$P = 40.27 \text{ cm}$$

$$P = 57.12 \text{ cm}$$

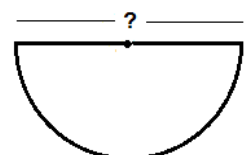
$$P = 24.90 \text{ m}$$



Assume sector with
internal angle of 270°
i.e. $\frac{3}{4}$ of a circle



Assume two $\frac{3}{4}$ circles



Length of curve = 15.23 m

Work backwards to
determine diameter

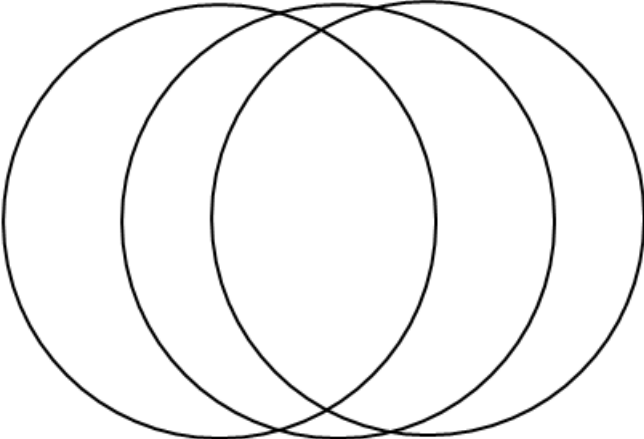
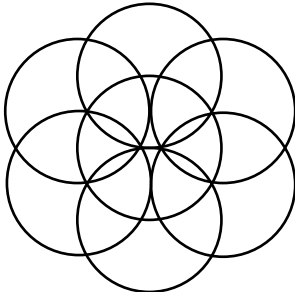
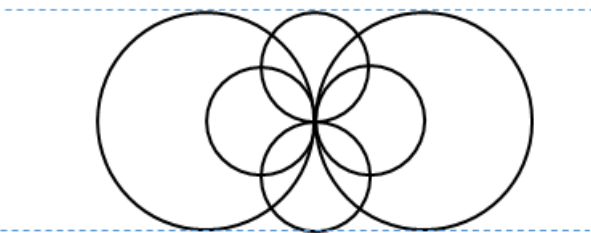
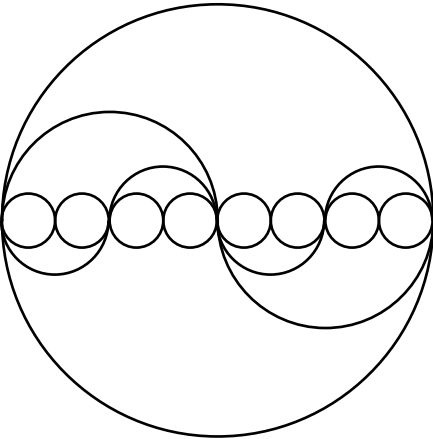
Jamos the Jeweller charges clients based on the amount of precious metal wire used in the jewellery designs. A recent client wants some custom pieces made and Jamos is in a bit of jam regarding these intricate circular designs.

Can you determine what length of the precious metal wire is in each design? What would the precious metal cost? Suggest an appropriate price to quote the client.

Note: Jamos charges clients \$25 per linear centimetre of the precious metal wire used.

Please present your calculations as clearly as possible so that Jamos may check your figures.

(Note that if one circle looks to have double or half of the diameter of another, it does.)

 Each circle has a diameter of 8.8 cm	 Each circle has a radius of 2.3 cm
 The largest circle has a radius of 3.6 cm	 The smallest circle has diameter 1.1 cm
BONUS QUESTION: What do you notice about the first and fourth designs on this page? Explain why this happens. Hint: express each length using algebra, using d for diameter of the large circle.	

Jamos the Jeweller charges clients based on the amount of precious metal wire used in the jewellery designs. A customer has provided design briefs that allow Jamos to create new designs. Can you help?

To make each design Jamos needs to know the diameter of each circle used.

Note: Jamos charges clients \$25 per linear centimetre of the precious metal wire used.

The largest pair of identical plain circular earrings with an overall cost of metal under \$500	Six circles, with the largest circumference measuring exactly 10cm, and the total cost of metal under \$625
A symmetrical design, with at least three different sized circles and metal cost between \$600 and \$650.	Eight interlocking circles, each of different size, with the largest diameter measuring no more than 12cm and the smallest diameter at least 1cm. Cost of metal between \$1000 and \$1500.